SOURCE: MEPC.402(83).pdf
emission values from marine diesel engines, as well as documentation and verification of CH4 and/or N2O emission values.
1.2 The measurements, calculations and reporting for CH 4 and/or N2O emission values should be carried out in accordance with the NO x Technical Code 2008 as amended, (NTC 2008) other than as specifically provided for in the protocol set out in appendix 1 of these Guidelines. All references in appendix 1 are to NTC 2008.
1.3 For onboard measurements, the protocol set out in appendix 1 may be accepted for an Individual Engine or for an Engine Group represented by the Parent Engine. It should not be accepted for an Engine Family without further justifications. For test-bed measurements, the protocol may also be accepted for an Engine Family.1
2 Information to be included in the CH4 and/or N2O file
2.1 The applicant for the establishment of the emission values should prepare a CH 4 and/or N2O file that should contain the following information:
.1 details of the engine as tested should include but are not limited to:
.1 model and designation;
.2 rated power and rated speed;
.3 listing of NO x critical components as fitted and settings / operating

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SOURCE: MEPC.402(83).pdf
.1 details of the engine as tested should include but are not limited to:
.1 model and designation;
.2 rated power and rated speed;
.3 listing of NO x critical components as fitted and settings / operating values as applied including, for CH 4, NO x certified maximum liquid-to-gas fuel ratios across load range; and
.4 other components and settings / operating values which affect CH4 and/or N2O emissions;
.2 details, including drawings of exhaust system, showing sampling position(s);
.3 where C slip is reported, including crankcase emissions, details of how that was determined should be provided;
.4 a copy of the relevant engine test data, as given in appendix 2 of these Guidelines and any additional data to fully define the engine performance and enable calculation of the gaseous emissions of CH4 and/or N2O. For test-bed measurements, this information can also be provided in the test report as referred to in section 5.10 of the NTC 2008;
1 The composition of an Engine Family or an Engine Group as defined in NTC 2008 is set solely by factors affecting NOx emissions; those cannot be assumed to be equally applicable to CH 4 and/or N2O emissions.

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SOURCE: MEPC.402(83).pdf
1 The composition of an Engine Family or an Engine Group as defined in NTC 2008 is set solely by factors affecting NOx emissions; those cannot be assumed to be equally applicable to CH 4 and/or N2O emissions. In some instances, such as liquid-to-gas fuel ratio, the requirements for highest NOx will tend to result in lowest CH4.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4) AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
Annex 7, page 3
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.5 where a CH4 and/or N2O reducing device or system is used, the CH4 and/or

N2O file should contain documentation on the emission abatement device, giving details of its intended purpose, manner of operation, critical components and settings / operating values together with information on any consumables necessary for its operation. Where exhaust gas samples are to be drawn from both before and after the device, details of analyser changeover arrangements and sequencing should be given if only one set of analysers is to be used.
3 Verification of the CH4 and/or N2O emission values

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SOURCE: MEPC.402(83).pdf
to be drawn from both before and after the device, details of analyser changeover arrangements and sequencing should be given if only one set of analysers is to be used.
3 Verification of the CH4 and/or N2O emission values
3.1 In order to confirm that the emission values have been established in accordance with these Guidelines, the CH 4 and/or N 2O file should be submitted to the Administration for verification.
3.2 On receipt of the CH 4 and/or N2O file from the applicant and satisfactory completion of the verification, a Statement of emission values for CH 4 and/or N2O should be issued by the Administration. The form of Statement of emission values for CH4 and/or N2O is set out in appendix 3 of these Guidelines.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4) AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
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APPENDIX 1
PROTOCOL FOR TEST-BED AND ONBOARD MEASUREMENTS OF CH4 AND/OR N2O EMISSIONS FROM MARINE DIESEL ENGINES BASED ON THE NOx TECHNICAL CODE 2008
Explanatory note:

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APPENDIX 1
PROTOCOL FOR TEST-BED AND ONBOARD MEASUREMENTS OF CH4 AND/OR N2O EMISSIONS FROM MARINE DIESEL ENGINES BASED ON THE NOx TECHNICAL CODE 2008
Explanatory note:
This protocol does not amend mandatory provisions in NTC 2008. Measurements, calculations and reporting of CH 4 and/or N2O emission values should be carried out in accordance with NTC 2008, other than as specifically provided for in the protocol.
Paragraph of NOx Technical Code 2008
Addition to NTC 2008
Abbreviations, subscripts and symbols
In table 1, the definitions of N2O and NMHC are added as follows:

Symbol Definition
N2O Nitrous oxide
NMHC Non-methane
hydrocarbons
In table 2, abbreviations for FTIR, NDUV, NMC are added as
follows:
Symbol Definition
FTIR Fourier transform infrared
(analyser)
NDUV Non-dispersive ultraviolet
(analyser)
NMC Non-methane cutter
Note: NMC = FID with non-methane cutter
In table 3, symbols and terms for CfCH4, CfN2O and Cslip-CH4 are
added as follows:
Symbol Term
CfCH4 g CH4 / g fuel2
CfN2O g N2O / g total fuel (applies to both gas
and liquid fuels)
Cslip-CH4 % (of the mass of the methane

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SOURCE: MEPC.402(83).pdf
In table 3, symbols and terms for CfCH4, CfN2O and Cslip-CH4 are
added as follows:
Symbol Term
CfCH4 g CH4 / g fuel2
CfN2O g N2O / g total fuel (applies to both gas
and liquid fuels)
Cslip-CH4 % (of the mass of the methane
containing fuel used by the energy
converter)
Note: Cslip-CH4 is a factor accounting for CH 4 (expressed in % of
mass of methane containing fuel consumed in the energy
converter) which is emitted from the energy converter (including
fuel from combustion chamber/oxidation process and from
crankcase, as appropriate).
2 For methane containing fuels, the Cslip-CH4 is covering the role of CfCH4, so CfCH4 is set to zero for these fuels.
For the purpose of these Guidelines, non-methane gas fuels should be regarded as liquid fuels.
RESOLUTION MEPC.402(83) (adopted on 11 April 2025)
GUIDELINES FOR TEST-BED AND ONBOARD MEASUREMENTS OF METHANE (CH4)
AND/OR NITROUS OXIDE (N2O) EMISSIONS FROM MARINE DIESEL ENGINESMEPC 83/17/Add.1
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Chapter 3
3.2 Test cycles and
weighting factors to be
applied
For the test cycles E2 and E3, the specific emission at the 10%

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Chapter 3
3.2 Test cycles and weighting factors to be applied

For the test cycles E2 and E3, the specific emission at the 10% mode point or the lowest mode point at which gas fuel would be used should be measured and reported in addition to the existing mode points.

For onboard measurements only:

In setting the load points of the test cycle to be followed the provisions of 6.4.6.7 should apply. In the case of the 100% load point this should, subject to the engine emission test plan, be allowed to be no lower than 85% of rated power. If that value cannot be achieved, then the test should be deferred to such time that at least that power level can be achieved. The test cycle 100% power weighting factor under 3.2 should be applied irrespective of the actual power developed at that load point.

Chapter 5
5.2.5.3 Positioning of abatement device or system